

A POMDP-Based Framework for Healthcare Decision-Making to Mitigate Cognitive Decline in Alzheimer's Disease

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Abstract—Alzheimer's disease staging involves sequential decisions from noisy clinical evidence. We formulate this process as a Partially Observable Markov Decision Process (POMDP), combining components estimated from the Alzheimer's Disease Neuroimaging Initiative (ADNI) with explicit modeling assumptions. The contributions are a benchmark with traceable parameter provenance, a comparison of four policy classes, and interpretation of the offline policy through belief-space decision regions. Across 100,000 simulated episodes per policy, SARSOP achieves the highest mild cognitive impairment (MCI) intake-stage agreement (81.2%) and lowest testing expenditure among informed policies. MyopicPOMDP achieves the highest Dementia intake-stage agreement (87.3%) without offline planning. Their episode-weighted overall accuracies are 78.3% and 78.1%, respectively, while equal-stage macro accuracies are 76.6% and 79.8%. Reward comparisons are conditional on the stopping protocol. Transition dynamics, initial beliefs, and APOE4 likelihoods are ADNI-derived; Mini-Mental State Examination (MMSE), Clinical Dementia Rating (CDR) likelihoods, and utilities are hand-specified. The results characterize simulated management decisions. Biological mitigation of cognitive decline remains an intended application rather than a demonstrated outcome.

Index Terms—POMDP, Alzheimer's disease, ADNI, healthcare decision-making, clinical decision support, SARSOP, belief state, cognitive decline

I. INTRODUCTION

Global dementia prevalence was estimated at 57.4 million in 2019 and projected to reach 152.8 million by 2050 [1]. Alzheimer's disease (AD) is a major cause of dementia. We represent clinical status as Cognitively Normal (CN), Mild Cognitive Impairment (MCI), or Dementia. CN denotes age-expected cognitive performance; MCI denotes impairment without substantial loss of independence; Dementia denotes impairment that disrupts independent living. Monitoring MCI supports reassessment and management planning [2]. Available evidence includes the 30-point Mini-Mental State Examination (MMSE) cognitive screen [3], the Clinical Dementia Rating (CDR) functional assessment [4], and the APOE4 genetic risk marker [5]. These instruments provide different, imperfect information about clinical status.

Computational approaches support AD assessment through stage classification from acquired measurements [6] and pre-

diction of future conversion [7]. These tasks provide useful estimates, but a prediction alone does not determine which additional assessment to obtain or when the available evidence warrants a management decision. In a sequential workflow, another test incurs a cost and can change the preferred action, while premature commitment risks a stage-inappropriate decision. The problem addressed here is therefore not only to estimate clinical stage, but to decide whether further information is worth acquiring.

A Partially Observable Markov Decision Process (POMDP) represents this problem by separating the latent clinical state from its noisy observations [8], [9]. It maintains a probability distribution, or belief, over stages and updates that distribution as evidence arrives. The policy compares the expected utility of a management commitment with the value of additional testing [10], [11]. Testing cost, uncertainty, and the consequences of an incorrect commitment thus enter a common decision objective. This formulation can also inform future connected-health workflows with intermittent measurements [12]; the present study evaluates the decision model, not a deployed monitoring system.

Prior decision-theoretic studies examine preclinical AD screening [13], dementia subtype differentiation [14], and diagnostic and treatment choices in acute stroke [15]. We investigate a complementary combination of CN–MCI–Dementia staging, multi-test selection, and management commitment, asking how policy choice changes stage agreement, testing expenditure, and commitment timing. Anticipation refers to utility-dependent commitment under uncertainty rather than demonstrated prediction of future decline. Mitigation is the intended healthcare application, not an established treatment effect: management actions end the simulated episode and do not alter disease progression.

The study makes three contributions. First, it separates the initial belief, transition dynamics, and APOE4 likelihoods estimated from the Alzheimer's Disease Neuroimaging Initiative (ADNI) cohort from hand-specified MMSE/CDR likelihoods and decision utilities. Second, it compares Random, Expert, MyopicPOMDP, and SARSOP under a shared evaluation

protocol, exposing subgroup and cost trade-offs. Third, it extracts the SARSOP policy into inspectable belief regions to characterize commitment decisions. Project code is available at <https://github.com/PannnTastic/AnticipAD>.

II. RELATED WORK

A. Alzheimer's Disease and Clinical Staging

AD pathology can precede clinical symptoms by years [16], [17]. ADNI links repeated cognitive assessments, imaging, and genetic markers [18]–[20], supporting classification and conversion-prediction studies [6], [7]. Using MMSE/CDR to estimate likelihoods against diagnoses partly defined by those scores introduces incorporation bias. Hand-specified likelihoods expose this assumption but do not establish independent diagnostic validity.

B. Partially Observable Markov Decision Processes

A POMDP comprises hidden states $\mathcal{S}$, actions $\mathcal{A}$, observations $\mathcal{O}$, transition probabilities T , observation likelihoods Z , rewards R , and discount factor $\gamma \in [0, 1)$ [8], [21]. A belief distribution is updated after each action and observation, and a policy maps that belief to an action [11]. Exact planning is generally intractable [9]; point-based methods plan on selected beliefs [22], with SARSOP emphasizing reachable regions [23]. POMCP and DESPOT instead use online search [24], [25]. Our reported comparison uses two baselines, one-step lookahead, and offline point-based planning through POMDPs.jl [26].

C. Sequential Medical Decision Support

Decision-theoretic clinical systems include treatment planning and allocation models [10], [27]; more recent reinforcement learning studies examine test acquisition and stopping [28]–[30]. For AD, Önen Dumlü et al. [13] consider pre-clinical screening, while Navarro Posada [14] studies dementia subtype differentiation. Khatim et al. [15] integrate diagnostic observations and treatment decisions for stroke.

Our contribution combines chronic staging, multiple assessment channels, management commitments, and policy interpretation. It complements these prior formulations by examining how policy choice changes intake-stage agreement, testing expenditure, and commitment timing. This is an application and evaluation contribution; the underlying planning algorithms are established methods.

III. POMDP MODEL

We define the POMDP as $\langle \mathcal{S}, \mathcal{A}, \mathcal{O}, T, Z, R, \gamma, b_0 \rangle$ with discount factor $\gamma = 0.95$. A discount of 0.95 places strong weight on future rewards while penalizing unnecessary delays. The evaluation cap, $H = 20$ decision epochs, was chosen to match the maximum number of visit records per participant in the assembled ADNI dataset before listwise exclusion (15,836 records; 3,777 participants). After removing records missing diagnosis or MMSE, the maximum is 19; the cap retains the original dataset-based value of 20. This choice bounds testing by the longest available visit sequence, not a clinically optimal

duration or SARSOP's planning horizon. Each non-terminal test represents one modeled follow-up visit with one selected assessment. Because preprocessing pools consecutive visits without standardizing their elapsed time, an epoch has units of visits, not a calibrated six-month duration. Consequently, $H = 20$ does not establish a ten-year follow-up period. Sensitivity to alternative caps remains untested.

A. ADNI Calibration and State Space

The disease-progression parameters, the APOE4 likelihoods, and the initial belief are estimated from the ADNI 1/GO/2/3 merged cohort [18]–[20], whereas the MMSE and CDR likelihoods are hand-specified clinical models (Table I) rather than population-representative frequencies. The calibration followed a stepwise cleaning process.

We started by extracting visit diagnoses from the ADNIMERGE2 diagnosis field. We then excluded non-Alzheimer etiologies and the Significant Memory Concern category, and merged early MCI into MCI to align the labels with our three-stage state space. That filtering step yielded 15,836 visit records from 3,777 participants, with missing values coded as NA. We next removed records missing either the diagnosis or the MMSE score listwise, leaving 12,464 records for the prior and transition estimates. No imputation was applied. The ADNI-derived APOE4 likelihoods were computed from observed values only. For modeling, MMSE scores were grouped into Normal (>24), Moderate (20–24), and Low (<20), CDR scores into 0, 0.5, and ≥ 1 , and APOE4 into Negative, Heterozygous, and Homozygous.

The latent state space is $\mathcal{S} = \{\text{CN}, \text{MCI}, \text{Dementia}\}$. The initial belief is estimated from the visit-level stage frequencies in the 12,464 retained records:

$$b_0 = [0.385, 0.417, 0.198]. \quad (1)$$

In (1), the entries are dimensionless probabilities in CN, MCI, Dementia order. Each $b_0(s)$ lies in $[0, 1]$ (0–100%) and the entries sum to one; the smallest and largest specified entries are 0.198 and 0.417. These proportions correspond to 4,801 CN, 5,198 MCI, and 2,465 Dementia records from 3,713 unique participants, so b_0 represents the empirical stage mixture after listwise exclusion, not a patient-intake prior.

The six actions comprise three tests and three terminal commitments. *MMSE*, *CDR*, and *APOE4* incur testing costs and return categorical observations without ending the episode. For each test, the simulator advances the latent state by one visit transition, then samples the result conditional on the updated state. *Wait* commits to routine surveillance without active treatment, *TreatMCI* to structured monitoring and care planning, and *TreatDem* to dementia management. These commitments end the episode; their clinical consequences are not simulated.

B. Transition, Observation, and Reward Parameters

Table I reports the transition, observation, and reward parameters. Transition frequencies are estimated from consecutive same-participant records after sorting by visit date,

TABLE I
MODEL PARAMETERS. ALL PROBABILITIES ARE DIMENSIONLESS. T AND APOE4 ARE ADNI-DERIVED WITH STRUCTURAL CONSTRAINTS; MMSE, CDR, AND UTILITIES ARE HAND-SPECIFIED. $\gamma = 0.95$; $H = 20$.

Visit transitions	CN	MCI	Dementia
From CN	0.929	0.068	0.003
From MCI	0	0.892	0.108
From Dementia	0	0	1
Observation given stage	CN	MCI	Dementia
MMSE Normal (> 24)	0.85	0.55	0.20
MMSE Moderate (20–24)	0.13	0.35	0.45
MMSE Low (< 20)	0.02	0.10	0.35
CDR 0	0.95	0.25	0.05
CDR 0.5	0.04	0.70	0.30
CDR ≥ 1	0.01	0.05	0.65
APOE4 Negative	0.69	0.56	0.34
APOE4 Heterozygous	0.28	0.35	0.47
APOE4 Homozygous	0.03	0.09	0.19
Action utility (points)	CN	MCI	Dementia
Wait	+30	−10	−1000
MMSE	−5	−5	−5
CDR	−20	−20	−20
APOE4	−50	−50	−50
TreatMCI	−50	+150	−50
TreatDem	−500	−100	+300

without interval filtering or time rescaling. Dementia is then made absorbing and MCI-to-CN reversion is fixed at zero. Only the selected test produces an informative observation, which means that the actions Wait, TreatMCI, and TreatDem return a null observation. The APOE4 likelihoods are estimated from ADNI frequencies, whereas the MMSE and CDR likelihoods are hand-specified clinical noise models, because diagnostic labels partly incorporate these instruments. Their original descriptions motivate cognitive and functional assessment [3], [4]; they do not establish the specific likelihood values in Table I. CDR achieves a 90-percentage-point separation between CN and Dementia (95% vs. 5% for CDR=0), compared with 65 percentage points for MMSE (85% vs. 20% Normal), making CDR more discriminative under these assumptions. APOE4 encodes genetic risk rather than direct staging evidence. The reward function encodes a clinical preference ordering: the −1000 penalty for waiting in Dementia captures the cost of failing to recognize advanced disease, while the −500 penalty for TreatDem in a CN patient guards against overtreatment. These dimensionless utility points express decision preferences, not monetary costs or clinically validated treatment benefits. Single-action rewards range from −1000 to +300. Sensitivity checks in Section V-C assess selected assumptions; they do not calibrate these utilities.

C. Belief Update

After an action a produces observation o , the belief is updated by Bayes’ rule:

$$b'(s') = \eta Z(o | s', a) \sum_s T(s' | s, a) b(s), \quad (2)$$

In (2), $s, s' \in \mathcal{S}$ denote the current and next latent stages, $a \in \mathcal{A}$ the selected action, and $o \in \mathcal{O}$ its categorical result. The probabilities $b(s)$ and $b'(s')$ describe the prior and updated beliefs. $T(s' | s, a)$ is the one-interval transition

probability and $Z(o | s', a)$ the observation likelihood. The normalization is $\eta = 1/P(o | b, a)$, where $P(o | b, a) = \sum_{s'} Z(o | s', a) \sum_s T(s' | s, a) b(s) > 0$. All probabilities are dimensionless, lie in $[0, 1]$, and normalize over their respective outcomes. Policies use this same update rule; their chosen tests and observed histories can nevertheless differ.

IV. POLICIES AND EVALUATION PROTOCOL

Random samples uniformly from the six actions and serves as an uninformed baseline. **Expert** applies fixed clinical thresholds ($p_{\text{CN}} \geq 0.80$, $p_{\text{MCI}} \geq 0.80$, $p_{\text{Dem}} \geq 0.85$) and otherwise tests in a fixed MMSE–CDR–APOE4 order. **MyopicPOMDP** performs a one-step Bayesian lookahead without training. For a candidate test a_{test} , its stage-independent immediate expected reward is

$$C(b, a_{\text{test}}) = \sum_s b(s) R(s, a_{\text{test}}), \quad (3)$$

and its discounted expected terminal value is

$$V(b, a_{\text{test}}) = \gamma \sum_o P(o | b, a_{\text{test}}) \times \max_{a' \in \mathcal{A}_{\text{term}}} \sum_{s'} b'_o(s') R(s', a'). \quad (4)$$

Here C in (3) is the immediate expected test reward, V in (4) is the discounted expected terminal reward, a_{test} is a candidate test, and a' is a candidate terminal action. $R(s, a)$ is the utility in Table I; $\gamma = 0.95$ is dimensionless. The observation probability is defined after (2), b'_o is its updated belief, and $\mathcal{A}_{\text{term}} = \{\text{Wait}, \text{TreatMCI}, \text{TreatDem}\}$ is the set of terminal commitment actions. The total one-step value of the test action is then the sum of the two terms,

$$Q(b, a_{\text{test}}) = C(b, a_{\text{test}}) + V(b, a_{\text{test}}), \quad (5)$$

The test value Q in (5) is compared with the immediate terminal values $Q(b, a') = \sum_s b(s) R(s, a')$; the maximizing action is selected. C , V , and Q are measured in the same utility points.

SARSOP [23] is an offline point-based solver that precomputes α -vectors over the reachable belief space and selects actions by value lookup during simulation. All experiments are implemented in Julia using the POMDPs.jl ecosystem [26].

Each solver is evaluated across ten random seeds with $N = 10,000$ episodes per seed, for 100,000 episodes per policy. An episode starts with a latent intake stage sampled from b_0 . Test actions update the stage and belief; a terminal action ends the episode. After $H = 20$ non-terminal tests, the implementation appends a fallback Wait commitment, so the recorded action count can reach 21. SARSOP training required approximately 300 s (requested precision 10^{-2} , timeout 300 s); this requested tolerance does not itself certify optimality. The stored policy was planned on a continuing model, whereas evaluation stops at commitment, which limits optimality-based comparisons with MyopicPOMDP.

For each seed, let n_s be the number of episodes with intake stage s , c_s the number whose terminal action matches that stage, and $N = \sum_s n_s = 10,000$. The reported metrics are

$$\begin{aligned} \text{Acc}_s &= 100 c_s / n_s, \\ \text{Acc}_{\text{overall}} &= 100 \sum_s c_s / N, \\ \text{Acc}_{\text{macro}} &= \frac{1}{3} \sum_s \text{Acc}_s. \end{aligned} \quad (6)$$

All accuracies in (6) range from 0 to 100%. Overall accuracy weights stages by their realized episode counts; macro accuracy weights the three stages equally. Table II reports means and sample standard deviations across ten seeds.

Correct intake-stage commitments are Wait for CN, TreatMCI for MCI, and TreatDem for Dementia. Fixing the intake label preserves a common reference across policies with different stopping times. This endpoint measures agreement with the presenting stage, not the appropriateness of management at commitment: an initially MCI case that progresses to Dementia and receives TreatDem is scored incorrect against intake. Consequently, these metrics cannot establish contemporaneous diagnostic accuracy or treatment efficacy. Reward uses the evolving pre-action state and a discount of γ^{t-1} at decision t , including a fallback commitment when needed. Testing cost sums undiscounted test utilities; mean decisions counts tests and the terminal action.

V. RESULTS

A. Benchmark Performance

Fig. 1 shows the qualitative behavior on representative patients, where colors encode actions: Wait (green), MMSE (yellow), CDR (blue), APOE4 (purple), TreatMCI (orange), and TreatDem (red); blank cells indicate epochs after the episode has ended. These selected examples illustrate short test sequences and terminal commitments; Random also terminates but can select an inappropriate action.

Table II reports the main results using episode-weighted overall accuracy from (6). Both belief-driven policies improve overall intake-stage accuracy and reward over Expert and Random, but do not dominate every subgroup or efficiency measure. SARSOP has lower Dementia accuracy than Expert, while Random has the lowest testing expenditure and fewest decisions. Equal stage weighting gives macro accuracies of $76.6 \pm 0.5\%$ (SARSOP), $79.8 \pm 0.3\%$ (MyopicPOMDP), $72.1 \pm 0.5\%$ (Expert), and $33.4 \pm 0.6\%$ (Random), computed per seed before aggregation. Across the ten shared seeds, every headline contrast is significant under a two-sided paired Wilcoxon signed-rank test with Holm correction ($p < 0.05$): SARSOP exceeds MyopicPOMDP in MCI accuracy by 9.8 pp, MyopicPOMDP exceeds SARSOP in Dementia accuracy by 19.1 pp, and MyopicPOMDP exceeds SARSOP in aggregate reward by 4.3 (95% CI [3.8, 4.8], Cliff's $\delta=1.0$). The reward advantage is partly an effect of the terminate-at-first-commitment protocol; the ranking is not a single-seed artifact. Since each seed already aggregates $N = 10,000$ episodes, this

(a) Intake stage: CN

Random	APOE4	Dem			
Expert	MMSE	CDR	Wait		
MyopicPOMDP	CDR	CDR	Wait		
SARSOP	CDR	CDR	Wait		
Decision epoch (visits)	1	2	3	4	5

(b) Intake stage: MCI

Random	MCI				
Expert	MMSE	CDR	APOE4	MCI	
MyopicPOMDP	CDR	CDR	MCI		
SARSOP	CDR	CDR	MCI		
Decision epoch (visits)	1	2	3	4	5

(c) Intake stage: Dementia

Random	APOE4	APOE4	MMSE	MCI	
Expert	MMSE	CDR	Dem		
MyopicPOMDP	CDR	CDR	CDR	Dem	
SARSOP	CDR	CDR	CDR	Dem	
Decision epoch (visits)	1	2	3	4	5

Fig. 1. Illustrative action sequences for simulated intake stages CN, MCI, and Dementia. The x-axis counts decision visits, starting at intake ($t = 1$); visits are not calibrated calendar intervals. Blank cells follow termination. Only the first five epochs are shown because all displayed sequences terminate by epoch 4.

test assesses the *stability* of the ranking across seeds rather than a clinical effect size, which would require outcome data external to the simulation.

Fig. 2 separates reward by intake stage. The contrasting MCI and Dementia distributions complement the intake accuracies in Table II; aggregate reward alone conceals these subgroup differences.

Fig. 3 illustrates policy-dependent belief histories. The MCI analysis gives 2.3 versus 3.5 mean decisions for SARSOP and Expert, compared with 2.50 versus 3.34 across all stages in Table II. This is consistent with earlier commitment under a lower utility-dependent boundary. Shared Bayesian equations do not imply identical observation histories: chosen tests and stopping influence the plotted averages. Reward affects belief indirectly through those choices.

B. SARSOP Policy Extraction

We evaluated the SARSOP α -vector policy on a uniform belief simplex grid and grouped points by the selected action (Table III, Fig. 4). This makes the policy's decision regions inspectable.

CDR is selected across 79.2% of the belief space as the most cost-effective discriminative test, while MMSE and APOE4 are never selected: under the hand-specified MMSE model the value of information from MMSE does not justify its cost relative to CDR, and APOE4 supplies genetic risk rather than staging evidence. The extracted commitment thresholds,

TABLE II

SOLVER PERFORMANCE (MEAN $\pm$ SD ACROSS 10 SEEDS, 10,000 EPISODES EACH). ACCURACIES SCORE INTAKE-STAGE AGREEMENT; OVERALL ACCURACY IS EPISODE-WEIGHTED. REWARD AND TEST COST ARE UTILITY POINTS. BOLD MARKS THE BEST COLUMN VALUE, INCLUDING THE SMALLEST DECISION COUNT AND LEAST NEGATIVE TESTING COST.

Policy	Reward	CN Acc. (%)	MCI Acc. (%)	Dem. Acc. (%)	Overall Acc. (%)	Cumul. Test Cost	Mean Decisions
SARSOP	68.7 $\pm$ 1.5	80.4 $\pm$ 0.7	81.2$\pm$0.6	68.2 $\pm$ 1.3	78.3$\pm$0.4	-29.9 $\pm$ 0.1	2.50 $\pm$ 0.01
MyopicPOMDP	73.0$\pm$1.2	80.5$\pm$0.4	71.4 $\pm$ 0.6	87.3$\pm$0.7	78.1 $\pm$ 0.3	-37.3 $\pm$ 0.1	2.87 $\pm$ 0.01
Expert	44.7 $\pm$ 1.9	79.8 $\pm$ 0.6	61.3 $\pm$ 0.5	75.1 $\pm$ 1.0	71.1 $\pm$ 0.4	-42.1 $\pm$ 0.4	3.34 $\pm$ 0.01
Random	-134.9 $\pm$ 4.4	33.5 $\pm$ 0.6	33.3 $\pm$ 0.8	33.3 $\pm$ 1.1	33.4 $\pm$ 0.5	-25.1$\pm$0.4	2.01$\pm$0.01

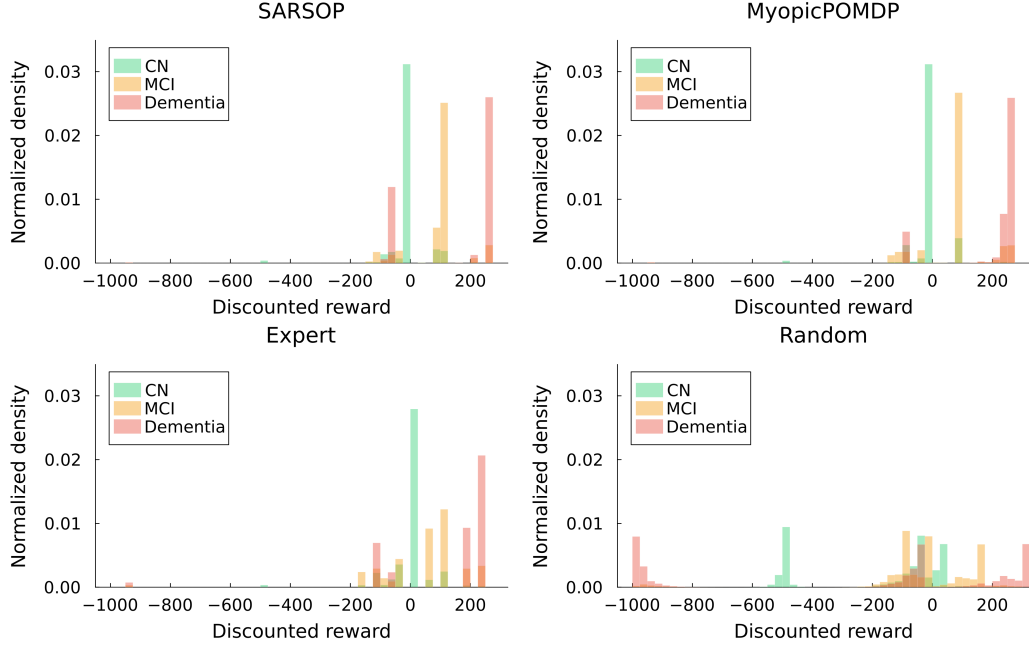


Fig. 2. Discounted reward distributions by intake stage and policy. The x-axis is in dimensionless utility points; density on the y-axis is per utility point.

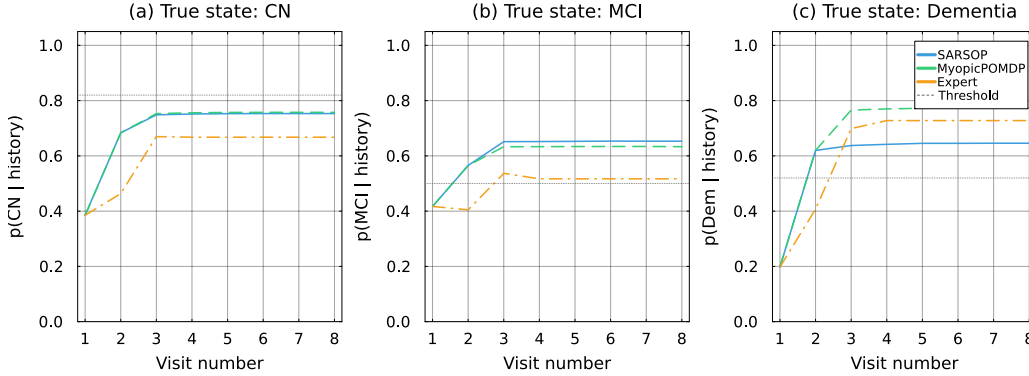


Fig. 3. Average target-state belief probability (dimensionless, 0-1) across decision visits ($N = 3,000$ episodes per group).

0.50 for MCI and 0.52 for Dementia, are far below Expert's fixed 0.80 and 0.85, which reflects the asymmetric reward: the -1000 cost of failing to treat Dementia favors acting on moderate certainty. Importantly, at the 0.50 MCI threshold the CN accuracy ($80.4 \pm 0.7\%$) remains close to Expert's ($79.8 \pm 0.6\%$), so these results show no increase in total CN intake-stage errors in this simulation. The minimum beliefs summarize the sampled regions; they are not sufficient one-

dimensional rules independent of the other beliefs.

C. Sensitivity Analysis

The archived reward experiment used 1,000 episodes, seed 42, and a five-decision cap, with SARSOP re-solved for each reward setting (requested precision 10^{-2} , timeout 60 s). Its baseline overall accuracies were 79.5% (SARSOP) and 79.0% (MyopicPOMDP). Changing the Dementia-Wait utility

TABLE III
SARSOP DECISION REGIONS: FRACTION OF THE BELIEF SIMPLEX
OCCUPIED PER ACTION AND MINIMUM BELIEF THRESHOLD FOR EACH
TERMINAL COMMITMENT.

Action	Belief (%)	Min-Belief Threshold
Test CDR	79.2	Dominant diagnostic action
Treat MCI	15.1	$p_{\text{MCI}} > 0.50$
Treat Dem.	5.1	$p_{\text{Dem}} > 0.52$
Wait	0.6	$p_{\text{CN}} > 0.82$
Test MMSE	0.0	Not selected on grid
Test APOE4	0.0	Not selected on grid

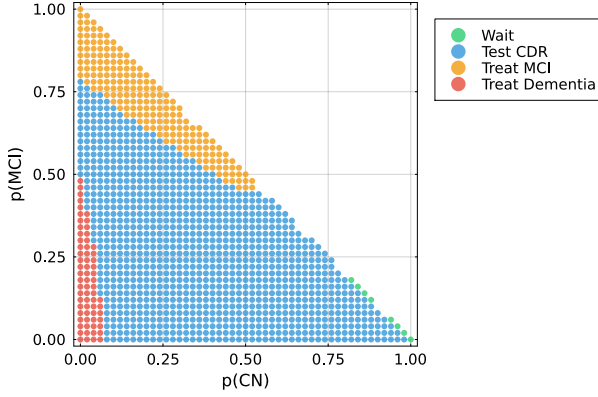


Fig. 4. SARSOP policy regions over the belief simplex (p_{CN} , p_{MCI} , p_{Dem}); CDR dominates the central diffuse region while terminal commitments occupy the belief corners.

from -1000 to -500 or -1500 left these accuracies unchanged to the reported precision, although rewards changed. Doubling test costs retained SARSOP's 79.5% but reduced MyopicPOMDP to 73.6%. With the MCI commitment reward reduced from 150 to 100, accuracy became 79.9% and 77.6%, respectively. Reward ordering was not invariant. These are separate, single-seed stress tests rather than confidence estimates for the main benchmark.

A further observation stress test used 10,000 episodes, seed 42, and the 20-decision cap. Replacing only the CDR MCI row (0.25, 0.70, 0.05) with empirical frequencies (0.075, 0.904, 0.021) and re-solving SARSOP gave overall accuracies of 81.6%, 82.2%, and 76.5% for SARSOP, MyopicPOMDP, and Expert. CDR remained selected at 77.9% of 231 sampled belief points. This supports CDR's prominence under that perturbation but also shows that the overall accuracy ordering can reverse. The check does not independently validate CDR or establish robustness to joint MMSE/CDR uncertainty.

VI. DISCUSSION

Transparent model integration (Contribution 1). Compared with prediction-focused AD models [6], [7], this framework makes test cost and stopping explicit. It complements preclinical screening [13] and subtype selection [14] by comparing management commitments across CN, MCI, and Dementia. Relative to the acute stroke formulation of [15], it studies a chronic, repeated-visit setting. Its strength is a traceable combination of empirical T , b_0 , and APOE4 with

openly stated MMSE/CDR and utility assumptions. CDR's dominance in Table III follows from the value of information under the specified test costs, not from any intrinsic superiority of CDR as an instrument.

Subgroup and efficiency trade-offs (Contribution 2). The choice between episode weighting and equal-stage weighting changes the accuracy comparison. SARSOP favors MCI intake recognition and lower test expenditure among informed policies, whereas MyopicPOMDP favors Dementia recognition. Because reward uses evolving states and accuracy uses intake labels, their rankings need not coincide. These results motivate reporting both subgroup agreement and decision costs; they do not establish a preferred policy for clinical deployment.

Interpretable commitment (Contribution 3). The extracted belief regions make the cost-information trade-off inspectable. Earlier MCI commitment (3.5 $\rightarrow$ 2.3 decisions in the illustrative analysis) is consistent with a lower utility-dependent commitment boundary, alongside policy-dependent test histories. The difference is 1.2 decision visits, not an observed calendar-time lead. The approach does not demonstrate mitigation of dementia: progression is action-independent, management actions terminate evaluation, and no post-treatment cognitive outcome is simulated. Mitigation in the title states the intended healthcare application, while the demonstrated contribution is a simulation framework for investigating management delay.

Limitations. Transitions are action-independent, so capturing treatment-induced disease slowing would require data from treatment arms. The benchmark combines ADNI-estimated and hand-specified components and evaluates them within the same simulator, so external validation on NACC [31] or OASIS [32] is needed before any clinical use. The population-level prior b_0 is applied uniformly to every patient; individualized priors could shift commitment timing. Intake-stage agreement can penalize appropriate responses to progression and needs comparison with decision-time staging. Transitions pool unequal visit intervals; same-day test bundles and elapsed-time-dependent progression are not modeled. The horizon cap and once-tested nature of APOE4 also need further assessment. Repeated APOE4 draws can overstate information. Neither same-model simulation nor single-seed sensitivity establishes clinical utility, observation-model validity, or general robustness.

VII. CONCLUSION

The framework combines ADNI-derived dynamics with explicit observation and utility assumptions, compares four policies, and exposes their belief-space decisions. Across 100,000 episodes per policy, SARSOP and MyopicPOMDP achieve 78.3% and 78.1% episode-weighted intake-stage accuracy, versus 76.6% and 79.8% macro accuracy. SARSOP favors MCI recognition and lower testing expenditure among informed policies; MyopicPOMDP favors Dementia recognition. These conditional simulation results support further study of management timing, but do not demonstrate slowed cognitive decline. Future work requires decision-time evaluation,

individualized priors, persistent genetic covariates, treatment effects, and external validation.

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